Transfer Portal Intelligence Platform
Model Documentation

Transfer Portal Intelligence Platform | Model methodology, interpretation, and decision-use guide

Prepared for basketball decision makers. This document explains the three production-facing models in plain language while preserving the important modeling details: what each model predicts, what data went into it, why the features were built, how the model was evaluated, and how the outputs should be used in the site.

Executive Summary

Model	Question answered	Primary output	Best use
Transfer Likelihood	How likely is this player to enter the portal?	Transfer probability, shown as a percent	Rank players by transfer risk and prioritize monitoring.
Transfer Movement	If he transfers, is he more likely to move up, down, or laterally?	Movement label plus numeric movement score	Identify candidates likely to reach stronger programs or fall to lower-level opportunities.
Movement-Based BPM	How could the player's BPM translate in a new landing context?	Projected next-season BPM and change from current BPM	Estimate future impact under likely, step-up, lateral, and step-down scenarios.

The system is designed as a decision-support layer, not an automated recruiting board. The strongest use is ranking, comparison, and question answering. A coach, front-office analyst, or roster decision maker can use the outputs to identify players worth deeper review, then pair the model output with film, scouting context, roster fit, academic/eligibility information, and live market intelligence.

Data Foundation

All three models begin from the same historical modeling table, `model_table_START`, built from public Bart Torvik player and team data. The historical window used for modeling is 2015 through 2025, with 2026 player rows used for current-site scoring. Player seasons are linked forward by player id and year to determine whether the player stayed, transferred, left the data, changed program level, or produced a next-season BPM.

Data group	Examples	Why it matters
Player production	BPM, OBPM, DBPM, PORPAG, ORTG, usage, points, rebounds, assists	Captures current ability, role, efficiency, and production shape.
Role and opportunity	Minutes, minutes share, team rank in minutes/BPM/usage, role label, class	Separates stars from low-role players and identifies underused players with room to grow.
Team context	Wins, conference wins, adjusted offense/defense, barthag, WAB, strength of schedule	Models whether the player is already succeeding in a strong environment or carrying a weaker one.
Conference/program level	High, upper-mid, and low tiers; program strength score	Creates a more realistic scale than conference name alone.

Engineered flags	Mid-major star, underused star, high-major role player, low-role high-output player	Turns basketball-specific hypotheses into structured model signals.

Model 1: Transfer Likelihood

The transfer likelihood model predicts whether a player who appears in one season is likely to appear at a different school in the next season. The output is a probability, not a yes/no verdict. On the site this is shown as a transfer percentage and is best interpreted as relative risk: a player at 45% should be monitored more aggressively than a similar player at 12%.

Target and model type

Item	Design choice
Target	Transfer next year: 1 if the player appears next season at a different school, 0 if he returns to the same school. Rows with no next-year player row are excluded from this specific model target.
Model	Logistic regression (`glm`, binomial family).
Train / validation / test split	Training through 2022, validation on 2023, test on 2024.
Operational threshold	0.33 was used for classification checks, while the site primarily uses the continuous probability.

Main feature logic

- Performance and role: games played, minutes, BPM, usage, ORTG, offensive rebounding rate, three-point percentage, PORPAG, and role label.
- Team and conference context: team wins, conference win percentage, barthag, adjusted offense, adjusted defense, and high-major vs. mid-major classification.
- Player-team fit: a player-team gap feature compares individual BPM to team strength, helping identify players who may be too productive for their current environment or buried on a stronger roster.
- Opportunity mismatch: minutes share, team rank features, and `bpm_minutes_gap` capture players whose performance may exceed their role.
- Interaction terms: MP x BPM, major x BPM, major x PORPAG, and player-team gap x opportunity gap allow the model to treat the same stat line differently depending on context.

Evaluation and interpretation

The model is evaluated with ROC/AUC, confusion matrices at the 0.33 threshold, and calibration bins comparing average predicted probability to actual transfer rate. The main decision value is ranking and calibration, not perfect binary classification. Transfer decisions are noisy because NIL, coaching changes, personal preference, academic factors, injuries, and roster promises are not in the public data.

Accuracy Assessment

I would rate this model as useful but moderate in pure accuracy, with its best value coming from ranking and risk-screening rather than hard yes/no classification. On the 2023 validation season it produced a 0.679 AUC, and on the 2024 test season it produced a 0.668 AUC, which means it separates likely transfers from likely stayers better than chance but is not close to perfect. At the 0.33 classification threshold, validation accuracy was 72.5% and test accuracy was 63.8%, but the balanced accuracy was only about 55% on validation and 54% on test, so I would not oversell it as a binary classifier.

The encouraging sign is calibration by rank: in validation, the lowest predicted decile transferred at about 8.7%, while the highest predicted decile transferred at about 51.1%. That is useful for a decision maker because it shows the model can sort players into meaningfully different risk groups. With more data, I would improve it with NIL signals, coaching changes, roster depth, scholarship availability, player recruiting pedigree, eligibility status, injury history, and in-season minutes trends.

Model 2: Transfer Movement

The movement model replaced the older destination conference model. Instead of asking only whether a player will land in a low-, mid-, or high-major tier, it estimates the size and direction of the player's likely program change. That makes the output easier to interpret: step up, lateral move, step down, or a major version of either direction.

Program strength score

A custom 0-100 program strength score is built for every team-season. It blends team quality and environment rather than relying on conference labels alone. This is what allows Duke to grade as clearly stronger than Boston College even though both are ACC schools, and allows a Duke-to-UNC move to read closer to lateral.

Component	Weight	Basketball meaning
Barthag	34%	Overall team quality / win probability strength.
WAB	18%	Resume strength and quality of wins/losses.
Adjusted offense	14%	How strong the team is offensively.
Adjusted defense	12%	How strong the team is defensively, reversed so stronger defense scores higher.
Strength of schedule	8%	Quality of competition.
Overall win percentage	7%	Actual results.
Conference win percentage	4%	League-level performance.
Conference tier score	3%	High / upper-mid / low context.

Target and model type

Item	Design choice
Target	Movement score = next program strength score minus current program strength score, only for historical players who actually transferred.
Model	Ridge regression using `glmnet` with 5-fold cross-validation.
Output labels	Major step up >= +25, step up +10 to +25, lateral -10 to +10, step down -25 to -10, major step down <= -25.
Site display	A clean label plus numeric score, for example `Step Up (+18.4)`.

Main feature logic

- Player quality and production: BPM, PORPAG, ORTG, usage, minutes, class, and role.
- Relative dominance: team and conference percentiles for BPM, usage, minutes, and PORPAG.

- Current environment: conference tier, team wins, conference win percentage, barthag, and current program strength.
- Basketball archetype flags: mid-major star, underused star, high-major role player, and low-role high-output player.
- Opportunity signals: minutes share and BPM-minus-minutes-percentile gap help identify players who may have more value than their current role shows.

Evaluation and interpretation

The model is evaluated as both a regression and a classification-style tool. Regression metrics include RMSE, MAE, and R-squared against the actual movement score. Basketball-facing checks include exact bucket accuracy, adjacent-bucket accuracy, and direction accuracy for up/lateral/down. The most important practical check is adjacent-bucket accuracy: being one bucket off is usually acceptable for scouting use, while confusing a major step up with a major step down would be a serious miss.

This model is more useful than the earlier destination conference model because it preserves magnitude. A player projected to move from Fairleigh Dickinson to Duke and a player projected to move from Boston College to Duke should not both be reduced to the same generic `high-major` class. The continuous score keeps those moves separate.

Accuracy Assessment

I would rate this as a strong and useful model for decision support, especially because it predicts direction and magnitude rather than only a broad destination tier. On held-out data, the movement score model had validation RMSE of 14.9 and MAE of 12.1 on a 0-100 program-strength scale, with R-squared of 0.606. On the 2024 test season it held up similarly with RMSE of 14.6, MAE of 11.8, and R-squared of 0.570. That means it explains a meaningful amount of real movement-score variation while keeping the average miss to roughly 12 program-strength points.

The most decision-useful accuracy check is direction and bucket closeness. The model was exactly right on the five movement buckets 46.2% of the time in validation and 43.1% in test, but it was within one bucket 92.2% of the time in validation and 93.4% in test. For the simpler up/lateral/down decision, it was 64.8% accurate in validation and 62.0% in test. With more data, I would improve it with roster needs by position, returning minutes, scholarship openings, geography, coaching style, prior recruiting rank, and known portal interest.

Model 3: Movement-Based BPM Projection

The BPM model projects next-season player impact after accounting for the kind of destination environment the player is likely to enter. It was rebuilt because the earlier destination-tier BPM model could behave counterintuitively, such as making an elite player look worse as the destination got easier. The new model predicts change in BPM, then adds that change to current BPM.

Target and model type

Item	Design choice
Target	Next-season BPM change: BPM_next minus current BPM.
Model	Elastic-net style `glmnet` model tuned across alpha values 0, 0.25, 0.50, 0.75, and 1 with 5-fold cross-validation.
Training population	Players with a next-season row, sufficient games and minutes, valid current and next BPM, and usable team/program context.

Scenario outputs	Likely movement path plus step-down, lateral, and step-up scenarios.

Main feature logic

- Current impact: BPM, OBPM, DBPM, PORPAG, ORTG, usage, minutes, points, rebounds, and assists.
- Current and next context: current program strength, projected next program strength, current program level, and next program level.
- Movement features: movement score, absolute movement size, up/down/lateral direction, and interaction terms between movement and BPM/usage/role.
- Role and dominance: minutes share, team percentiles, conference percentiles, class, and role label.
- Manual context adjustment: elite and conference-dominant players receive a step-down boost when moving to easier contexts and a step-up drag when moving to harder contexts, reflecting expected translation of role and opponent quality.

Evaluation and interpretation

The BPM model is evaluated on held-out validation and test seasons using next-BPM RMSE, MAE, and R-squared, along with separate metrics for predicted BPM change. It is also compared against simple baselines: assuming the player's BPM stays the same and assuming the average historical BPM change. Additional diagnostics split performance by movement direction, current program level, next program level, and current BPM bucket.

The output should be read as an impact estimate, not a guarantee. A projected BPM of +6.5 means the model views the player as likely to perform at roughly that statistical impact level in the given context. The change value is often the more decision-useful number because it tells whether the model expects the player's production to translate up, down, or roughly hold.

Accuracy Assessment

I would rate this as the strongest model numerically, while still needing guardrails because BPM itself can be volatile. On validation, the movement-based BPM model produced RMSE of 2.38 BPM, MAE of 1.85 BPM, and R-squared of 0.613. On the 2024 test season it was very similar: RMSE of 2.41, MAE of 1.87, and R-squared of 0.606. In plain English, the model is typically missing next-season BPM by just under two BPM points on average, which is a useful level of accuracy for screening and comparison.

It also beats simple baselines. On the test season, assuming a player's BPM simply stays the same had RMSE of 2.98 and MAE of 2.29, while the new movement-based model improved that to RMSE of 2.41 and MAE of 1.87. That is a meaningful gain. With more data, I would improve it with projected minutes, returning team usage, lineup context, shot quality, play type, defensive assignment difficulty, team style, injury history, and scouting grades, while also reviewing extreme projected BPM outputs before high-stakes use.

How Accuracy Should Be Read

Metric	Used for	Plain-English interpretation
AUC	Transfer likelihood	How well the model separates likely transfers from likely stayers across thresholds.
Confusion matrix	Transfer likelihood and movement labels	How many predictions are correct at a chosen threshold or label bucket.
Calibration bins	Transfer likelihood	Whether 20% predictions actually transfer about 20% of the time.

RMSE	Movement score and BPM	Typical error size, with larger misses punished more heavily.
MAE	Movement score and BPM	Average absolute miss; often easiest to explain to non-technical readers.
R-squared	Movement score and BPM	How much variation the model explains compared with predicting the mean.
Adjacent-bucket accuracy	Movement model	Whether the model is at least directionally close even when the exact label is off.

For basketball use, exact prediction is less important than stable ranking and directional correctness. The best workflow is to use the models to narrow the player universe, then review the highest-priority names with film, scouting notes, roster construction, eligibility, and market context.

Current Site Outputs

Output	Where it appears	Decision-maker reading
Transfer %	Model projections, player table, player pages, Portal Pulse	How aggressively the player should be monitored as a portal risk or potential target.
Movement label and score	Model projections, player table, player pages, Portal Pulse	Likely destination direction and how large the move is expected to be.
Projected BPM and change	Model projections, player table, player pages	Expected statistical impact in the next context and whether that impact rises or falls from current level.
Scenario BPM	Model projections	How the player's future BPM changes under step-down, lateral, and step-up landing assumptions.

Strengths, Limitations, and Recommended Use

Strengths

- Uses player production, role, team strength, conference context, and engineered basketball features rather than relying on raw box-score stats alone.
- Separates transfer risk from destination movement and future performance, which makes the outputs easier to explain.
- The movement model gives a continuous scale, avoiding the over-simplification of destination conference tiers.
- The BPM model explicitly accounts for destination difficulty and role translation.

Limitations

- The data is public. It does not include NIL, private recruiting interest, academic status, family preference, injuries, coaching promises, or live roster needs.
- Historical labels depend on matching player ids across seasons; any public-data matching issue can affect training labels.
- Transfer movement and BPM projections can be noisy for unusual players, very low-minute players, and players with major role changes.
- The BPM model should be monitored for extreme outliers and may benefit from production guardrails or capped outputs before high-stakes use.

Recommended workflow

1. Use Portal Pulse or the model projection tables to identify high-transfer-probability players.
2. Filter those players by movement direction to find players likely to move up, move down, or remain lateral.
3. Review projected BPM and BPM change to understand whether the player is expected to translate statistically.
4. Use player pages to inspect profile, role, production shape, and similar-player context.
5. Move final decisions into a scouting workflow with film, roster fit, medical/eligibility checks, and human intel.

Implementation Notes

The modeling code lives in `Barttorvik_API.Rmd`. The final site-facing scored file is `data/complete_2026_scored.csv` and is copied to `docs/data/complete_2026_scored.csv` for GitHub Pages. The current site data contains 4,979 scored players and includes non-missing movement predictions for all 4,979 rows.

Model artifact / object	Purpose
`glm_fit`	Transfer likelihood logistic regression.
`movement_cv_fit` and `movement_meta`	Movement score ridge model and factor/design-matrix metadata.
`new_bpm_cv_fit` and `new_bpm_meta`	Movement-based BPM projection model and scoring metadata.
`score_transfer_model()`	Scores current players with transfer probability.
`score_movement_model()`	Scores current players with movement score/label/direction.
`score_new_bpm_model()`	Scores current players and scenario grids with projected BPM and BPM change.

Bottom Line

The three-model structure gives TPIP a clear workflow: first identify who may move, then estimate the level of move, then estimate how the player's production may translate after the move. The outputs are not meant to replace basketball judgment. They are meant to make the first pass faster, more consistent, and more explainable.